

1 Appendix for Section 4

1.1 Apparent Balanced Growth in $\bar{c}$ and $\text{cov}(c, \mathbf{p})$

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the GIC, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\text{cov}(c, \mathbf{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\text{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant.

{sec:ApndxBalancedGrowth}

1.2 $\log c$ and $\log(\text{cov}(c, \mathbf{p}))$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over the last 1000 periods (of a 2000 period simulation).¹ The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\text{cov}(c, \mathbf{p}))$ also increases apparently linearly. (These results are produced by the notebook `ApndxBalancedGrowthcNrmAndCov.ipynb`).

¹For an exposition of our implementation of Harmenberg's method, see this supplemental appendix.

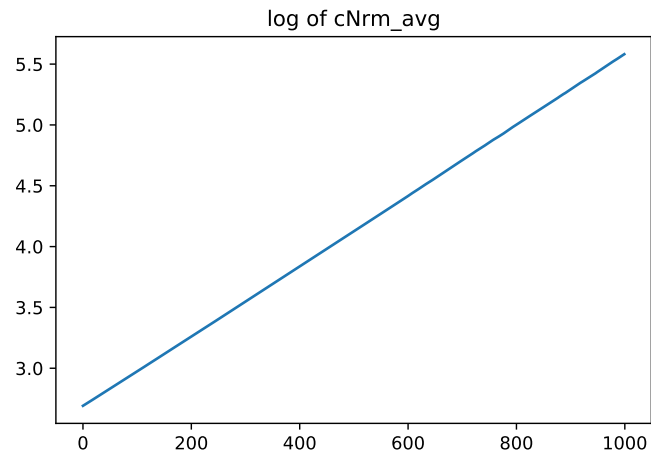


Figure 1 Appendix: $\log \mathfrak{c}$ Appears to Grow Linearly

{fig:logcNrm}

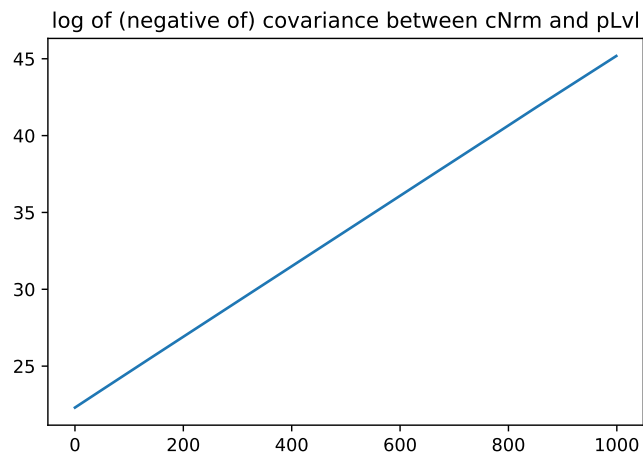


Figure 2 Appendix: $\log (-\text{cov}(c, \mathbf{p}))$ Appears to Grow Linearly

{fig:logcov}

(See (2))

Table 1 Model Characteristics Calculated from Parameters

Description	Symbol and Formula	Approximate Calculated Value
Finite Human Wealth Factor	$\tilde{\mathcal{R}}^{-1} \equiv \mathcal{G}/\mathbf{R}$	0.990
PF Value of Autarky Factor	$\sqsupset \equiv \beta \mathcal{G}^{1-\gamma}$	0.932
Growth Compensated Permanent Shock	$\underline{\psi} \equiv (\mathbb{E}[\psi^{-1}])^{-1}$	0.990
Uncertainty-Adjusted Growth	$\underline{\mathcal{G}} \equiv \mathcal{G} \underline{\psi}$	1.020
Utility Compensated Permanent Shock	$\underline{\underline{\psi}} \equiv (\mathbb{E}[\psi^{1-\gamma}])^{1/(1-\gamma)}$	0.990
Utility Compensated Growth	$\underline{\underline{\mathcal{G}}}_u \equiv \mathcal{G} \underline{\underline{\psi}}$	1.020
Absolute Patience Factor	$\mathbf{P} \equiv (\mathbf{R}\beta)^{1/\gamma}$	0.999
Return Patience Factor	$\mathbf{P}/\mathbf{R} \equiv \mathbf{P}/\mathbf{R}$	0.961
Growth Patience Factor	$\mathbf{P}/\mathcal{G} \equiv \mathbf{P}/\mathcal{G}$	0.970
Modified Growth Patience Factor	$\mathbf{P}/\mathcal{G}\mathbb{E}[\psi^{-1}] \equiv \mathbf{P}/\underline{\mathcal{G}}$	0.980
Value of Autarky Factor	$\sqsubseteq \equiv \beta \mathcal{G}^{1-\gamma} \underline{\underline{\psi}}^{1-\gamma}$	0.941
Weak Return Impatience Factor	$\wp^{1/\gamma} \mathbf{P} \equiv (\wp \beta \mathbf{R})^{1/\gamma}$	0.071

{table:Calibration}